APS502H1

Financial Engineering

Fall 2024

Computational Project Report

Present to

Professor Roy Kwon

By

Nara Cheyklin

ID: 1011835971

Problem 1 Part 1(a)

<u>Goal</u>: Minimize the total cost of the bond portfolio

Minimize $c' x$

<u>Constraint:</u> 1. The obligation of required cash flow in each year

$Ax >= b$

2. Each number of bonds must not be less than 0

$lb = zeros(12,1)$

<u>Variable:</u>

Let's c be the bond price vector $\begin{bmatrix} 109 \\ 94.8 \\ 99.5 \\ 93.1 \\ 97.2 \\ 96.3 \\ 92.9 \\ 110 \\ 104 \\ 101 \\ 107 \\ 102 \end{bmatrix}$ Let's x be the number of bond vector = $\begin{bmatrix} units\ of\ bond\ 1 \\ units\ of\ bond\ 2 \\ units\ of\ bond\ 3 \\ units\ of\ bond\ 4 \\ units\ of\ bond\ 5 \\ units\ of\ bond\ 6 \\ units\ of\ bond\ 7 \\ units\ of\ bond\ 8 \\ units\ of\ bond\ 9 \\ units\ of\ bond\ 10 \\ units\ of\ bond\ 11 \\ units\ of\ bond\ 12 \end{bmatrix}$

Let's A be the cash flow matrix= $\begin{bmatrix} 10 & 7 & 8 & 6 & 7 & 6 & 5 & 10 & 8 & 6 & 10 & 7 \\ 10 & 7 & 8 & 6 & 7 & 6 & 5 & 10 & 8 & 6 & 110 & 107 \\ 10 & 7 & 8 & 6 & 7 & 6 & 5 & 110 & 108 & 106 & 0 & 0 \\ 10 & 7 & 8 & 6 & 7 & 106 & 105 & 0 & 0 & 0 & 0 & 0 \\ 10 & 7 & 8 & 106 & 107 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 110 & 107 & 108 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Let's b be the obligation cash flow vector = $\begin{bmatrix} 500 \\ 200 \\ 800 \\ 200 \\ 800 \\ 1200 \end{bmatrix}$

Let's lb be the lower bound number of each bond zeros(12,1)

Methodology:

As in *linprog* function Ax<= b, Inputting -Ax<= -b ensures that Ax>=b

with MATLAB,

[x=the optimal number-of-bond vector, fval = the total cost of portfolio of the optimal solution]

= *linprog(c',-A,-b,[],[],lb,[])* ,then I obtained

$$x=\begin{bmatrix} 44.69 \\ 0 \\ 0 \\ 0 \\ 3.3 \\ 0 \\ 0 \\ 3 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ and fval= } 5522$$

See MATLAB code next page

Summary

The optimal solution is to have 44.69 units of bond 1, 3.3 units of bond 5, and 3 units of bond 8, and the total cost for the portfolio is 5522

Problem 1 Part 1 (b)

<u>Goal</u>: Minimize the total cost of the bond portfolio

Minimize c_new' x

<u>Constraint:</u> 1. The obligation of required cash flow in each year

Ax>= b

2. Each number of bonds must not be less than 0

lb = zeros(12,1)

3. Bond 2 and 3 must not be more than 5% of the total units

Units of bond 2<= 0.05total units and Units of bond 3 <= 0.05total units

<u>Additional Variables from Problem 1 part 1(a):</u>

Let's A_add1 be additional row constraint for bond 2 and bond 3 and total units=

$$\begin{bmatrix} 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.05 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.05 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 \end{bmatrix}$$

Let's A_add2 be additional column constraint for bond 2 and bond 3 and total units = zeros(6,1)

Let's b_add be additional bound = [0;0;0]

Let's c_new be c with additional 0 in the last row

$$\text{x_new}=\begin{bmatrix} units\ of\ bond\ 1 \\ units\ of\ bond\ 2 \\ units\ of\ bond\ 3 \\ units\ of\ bond\ 4 \\ units\ of\ bond\ 5 \\ units\ of\ bond\ 6 \\ units\ of\ bond\ 7 \\ units\ of\ bond\ 8 \\ units\ of\ bond\ 9 \\ units\ of\ bond\ 10 \\ units\ of\ bond\ 11 \\ units\ of\ bond\ 12 \\ total\ units \end{bmatrix}$$

<u>Methodology:</u>

with MATLAB,

[x_new=the optimal number-of-bond vector, fval_new = the total cost of portfolio of the optimal solution]

= *linprog(c_new',-[A, A_add2; A_add1],-[b;b_add],[],[],lb,[])* ,then I obtained

$$X_new=\begin{bmatrix} 44.69 \\ 0 \\ 0 \\ 0 \\ 3.3 \\ 0 \\ 0 \\ 3 \\ 0 \\ 0 \\ 0 \\ 0 \\ 50.99 \end{bmatrix} \text{ and } fval_new= 5522$$

See MATLAB code next page

Summary:

The optimal solution is to have 44.69 units of bond 1, 3.3 units of bond 5, and 3 units of bond 8, and the total cost for the portfolio is 5522. When comparing, to the result in (a), the units of bond 2 and bond 3 were already zero in the optimal solution, both results are the same.

Alternatively, if the objective were to find the optimal portfolio given that the units of bond 2 and 3 **must not be less than** 5% of the total units, the MATLAB can be altered by changing A_add1 to

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.05 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.05 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & 1 \end{bmatrix}$$

Problem 1 Part 2

Goal: Minimize the total cost of the bond portfolio

Minimize c'x

Constraint: 1. Each number of bonds must not be less than 0

lb = zeros(12,1)

2. The sum of (units of each bond) (its price) must match the present value of liabilities

3. The sum of (units of each bond) (each bond duration) must match the duration of liability

Additional Variables from Part 1:

Let Aeq be [[present value of each bond] ; [durations of each bond]] =

[[108.99544119416079, 94.7974149528402, 99.53009036661373, 93.10190581446297, 97.21789435767225, 96.36989564784616, 92.93796685018896, 109.85370079299102, 104.50540965491467, 99.15711851683834, 107.53071510623309, 102.00164575219686]

; [484.81772630028456, 441.7837348255197, 456.12839865044134, 382.60293221465054, 393.53388515906147, 329.37238347580234, 321.61160267687325, 283.75420557859707, 273.8887979543835, 264.0233903301699, 194.0936548208736, 186.3283584082791]]

Let beq be [present value of liabilities; duration of liability] =

[2757.9367548911464; 9863.211578452068]

Methodology:

With Python, I obtained the present value of liabilities, the duration of liabilities and the durations of bonds and input them into MATLAB.

With MATLAB,

[x=the optimal number-of-bond vector, fval = the total cost of portfolio of the optimal solution]

= *linprog(c',[],[],Aeq,beq,lb,[])* ,then I obtained

$$x = \begin{bmatrix} 0 \\ 0 \\ 13.4945 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 13.5384 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and fval} = 2750.7$$

See Python at the buttom and MATLAB code next page

<u>Summary:</u>

The optimal solution is to have 13.49945 units of bond 3 and 13.5384 units of bond 9, and the total cost for the portfolio is 2750.7. Compared to Part 1, the units of bonds are either 0 or almost identical to each other, while the units of bonds in part 1 are quite varied. The total cost of portfolio in this part is approximately half of the total cost in part 1 due to the difference in constraint, which reflects that portfolio in part 2 is less sensitive than part 1.

```python
liabilities=[500,200,800,200,800,1200]
spot=[0.0504, 0.0594, 0.0636, 0.0718, 0.0789, 0.0839]
CashFlow=[[10,10,10,10,10,110],[7,7,7,7,7,107],[8,8,8,8,8,108],[6,6,6,6,106,0],[7,7,7,7,107,0],[6,6,6,106,0,0],
          [5,5,5,105,0,0],[10,10,110,0,0,0],[8,8,108,0,0,0],[6,6,106,0,0,0],[10,110,0,0,0,0],[7,107,0,0,0,0]]
liab_Duration=0
bond_Duration=[]
NPVliab=0
NPVbond=[]
for i in range(6): #Calculate NPV and Duration of liabilities
    NPVliab+=liabilities[i]/(1+spot[i])**(i+1)
    liab_Duration+=((i+1)*liabilities[i])/((1+spot[i])**(i+2))

for c in CashFlow: #Calculate NPV and Duration of each bond
    di=0
    pi=0
    for i in range(6):
        pi+=c[i]/(1+spot[i])**(i+1)
        di+=(((i+1)*c[i])/((1+spot[i])**(i+2)))
    bond_Duration.append(di)
    NPVbond.append(pi)

print(liab_Duration) #9863.211578452068
print(NPVliab) #2757.9367548911464
print(bond_Duration) #[484.81772630028456, 441.7837348255197, 456.12839865044134, 382.60293221465054, 393.53388515906147, 329.37238347580234,
                     #321.61160267687325, 283.75420557859707, 273.8887979543835, 264.0233903301699, 194.0936548208736, 186.3283584082791]
print(NPVbond) #[108.99544119416079, 94.7974149528402, 99.53009036661373, 93.10190581446297, 97.21789435767225, 96.36989564784616,
               #92.93796685018896, 109.85370079299102, 104.50540965491467, 99.15711851683834, 107.53071510623309, 102.00164575219686]
```

Python Code for Problem 1 Part 1

Problem 2 Part 1 (a)

	SPY	GOVT	EEMV
Arithmatic Expected Return	0.01343	0.00056	0.00502
Geometric Expected Return	0.01241	0.00045	0.00437
Standard Deviation	0.04527	0.01486	0.036090

Covariance Table:

Covariance	SPY	GOVT	EEMV
SPY	0.00205	0.00012	0.00115
GOVT	0.00012	0.00022	0.00014
EEMV	0.00115	0.00014	0.00130

Problem 2 Part 1 (b)

Goal: Minimize the variance of ETFs

Minimize 0.5x' Q x

Constraint: 1. The expected return of portfolio must not be less than the expected goal return

Ax>=R

2. The sum of proportions of funds in a portfolio must be equal to 1

Aeq x= beq

Variables:

Let's x the vector of proportion of each fund in the portfolio = $\begin{bmatrix} Weight\ of\ SPY \\ Weight\ of\ GOVT \\ Weight\ of\ EEMV \end{bmatrix}$

Lets Q be the covariance matrix of portfolio = $\begin{bmatrix} 0.00205 & 0.00012 & 0.00115 \\ 0.00012 & 0.00022 & 0.00014 \\ 0.00115 & 0.00014 & 0.00130 \end{bmatrix}$

Let's c = zeros(3,1)

Let's A be - the matrix of the geometric expected return of funds= - $\begin{bmatrix} 0.00045 & 0.00437 & 0.00115 \end{bmatrix}$

Let's goal_R be the goal expected return matrix = [0.001:0.001:0.010]

Let's Aeq = ones(1,3) and beq be [1]

Methodology: (Short-selling is allowed)

For each expected goal return:

[x, fval] = quadprog(Q, c, A, -each expected goal return, Aeq, beq,[],[])

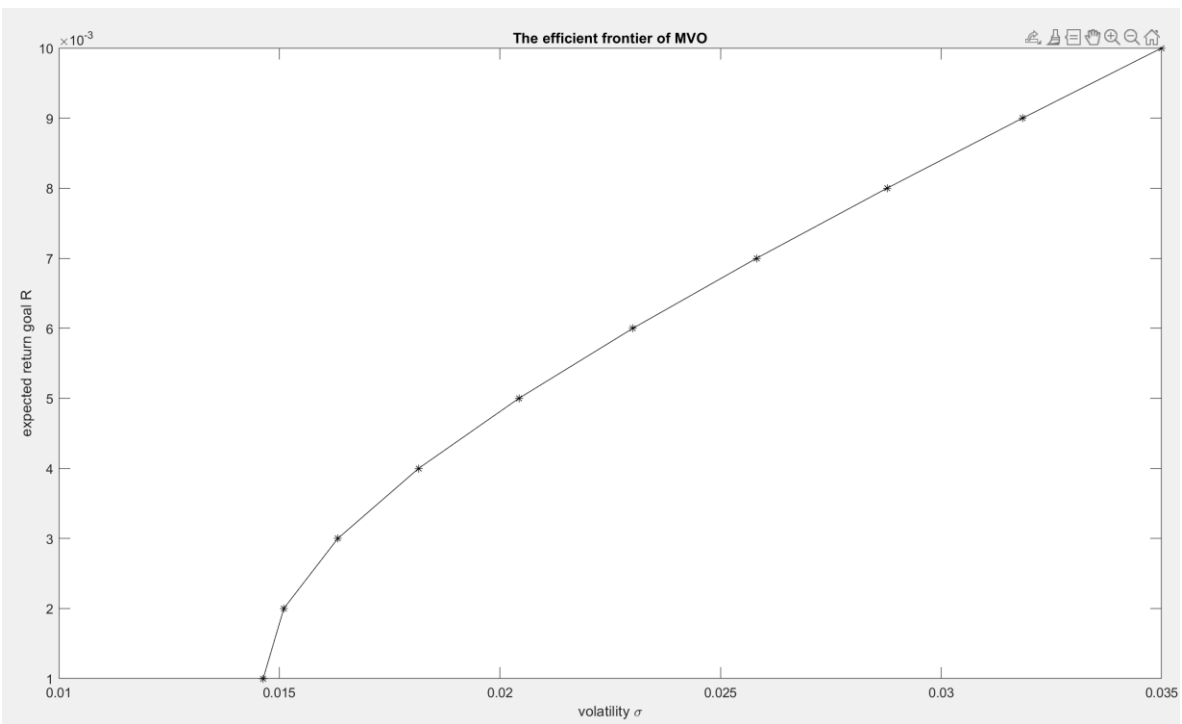
Standard deviation of each portfolio = (x Q x')

See MATLAB code next page

Summary:

These are the weight of each fund and portfolio volatility and variance for each expected goal return.

Expected Goal Return R	Weight of SPY	Weight of GOVT	Weight of EEMV	Portfolio Volatility	Portfolio Variance
0.10%	0.0342	0.9293	0.0366	0.0146	0.000213
0.20%	0.1343	0.8800	-0.0142	0.0151	0.000228
0.30%	0.2346	0.8306	-0.0651	0.0163	0.000266
0.40%	0.3349	0.7812	-0.1160	0.0182	0.000331
0.50%	0.4351	0.7318	-0.1669	0.0204	0.000416
0.60%	0.5354	0.6824	-0.2178	0.0230	0.000529
0.70%	0.6357	0.6330	-0.2687	0.0258	0.000666
0.80%	0.7360	0.5836	-0.3196	0.0288	0.000829
0.90%	0.8363	0.5342	-0.3705	0.0318	0.001011
1.00%	0.9366	0.4848	-0.4214	0.0350	0.001223



Efficient Frontier

Problem 2 Part 2

	CME	BR	CBOE	ICE	ACN
Arithmetic Expected Return	0.01345	0.01667	0.01454	0.01326	0.01497
Geometric Expected Return	0.01189	0.01460	0.01259	0.01146	0.01269
Standard Deviation	0.05657	0.06507	0.06223	0.06080	0.06775

Covariance Table:

	SPY	GOVT	EEMV	CME	BR	CBOE	ICE	ACN
SPY	0.00205	0.00012	0.00115	0.00090	0.00191	0.00104	0.00086	0.00233
GOVT	0.00012	0.00022	0.00014	0.000003	0.00023	0.00001	0.00005	0.00022
EEMV	0.00115	0.00014	0.00130	0.00041	0.00094	0.00047	0.00032	0.00117
CME	0.00090	0.000003	0.00041	0.00320	0.00112	0.00189	0.00117	0.00112
BR	0.00191	0.00023	0.00094	0.00112	0.00423	0.00123	0.00070	0.00288
CBOE	0.00104	0.00001	0.00047	0.00189	0.00123	0.00387	0.00151	0.00156
ICE	0.00086	0.00005	0.00032	0.00117	0.00070	0.00151	0.00370	0.00107
ACN	0.00233	0.00022	0.00117	0.00112	0.00288	0.00156	0.00107	0.00459

Goal: Minimize the variance of ETFs

Minimize $0.5x' Q x$

Constraint: 1. The expected return of portfolio must not be less than the expected goal return

$$Ax>=R$$

2. The sum of proportions of funds in a portfolio must be equal to 1

$$Aeq\ x= beq$$

Variables:

Let's Q be the covariance matrix of portfolio =
[0.00205,0.00012,0.00115,0.00090,0.00191,0.00104,0.00086,0.00233;
0.00012,0.00022,0.00014,0.000003,0.00023,0.00001,0.00005,0.00022;

0.00115,0.00014,0.00130,0.00041,0.00094,0.00047,0.00032,0.00117;

0.00090,0.000003,0.00041,0.00320,0.00112,0.00189,0.00117,0.00112;

0.00191,0.00023,0.00094,0.00112,0.00423,0.00123,0.00070,0.00288;

0.00104,0.00001,0.00047,0.00189,0.00123,0.00387,0.00151,0.00156;

0.00086,0.00005,0.00032,0.00117,0.00070,0.00151,0.00370,0.00107;

0.00233,0.00022,0.00117,0.00112,0.00288,0.00156,0.00107,0.00459]

Let's A be - the geometric expect rate of return =

[0.01241, 0.00045, 0.00437, 0.01189, 0.01460, 0.01259, 0.01146, 0.01269]

c=zeros(8,1), Aeq=ones(1,8)

For each expected goal return:

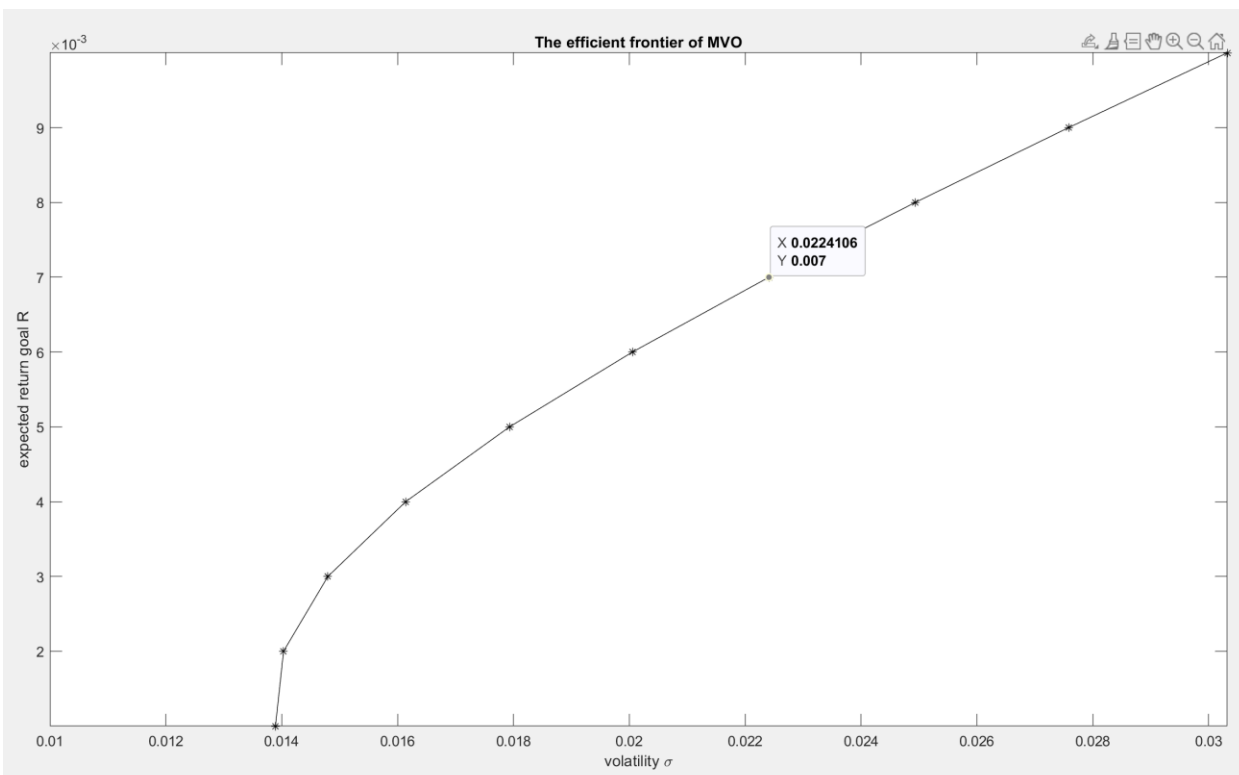
[x, fval] = quadprog(Q, c, A, -each expected goal return, Aeq, beq,[],[])

Standard deviation of each portfolio = (x Q x')

See MATLAB code next page

Summary:

Expected Goal Return	Weight SPY	Weight GOVT	Weight EEMV	Weight CME	Weight Br	Weight CBOE	Weight ICE	Weight ACN	Portfolio Volatility	Portfolio Variance
0.10%	0.0677	0.8944	0.0292	0.0411	-0.0294	0.0277	0.0168	-0.0475	0.0139	0.00019
0.20%	0.1091	0.8575	0.0066	0.0474	-0.0198	0.0322	0.0234	-0.0565	0.0140	0.00019
0.30%	0.1746	0.7993	-0.0291	0.0572	-0.0046	0.0393	0.0338	-0.0706	0.0148	0.00022
0.40%	0.2401	0.7410	-0.0648	0.0671	0.0107	0.0464	0.0443	-0.0847	0.0161	0.00026
0.50%	0.3710	0.6826	-0.1005	0.0769	0.0259	0.0536	0.0547	-0.0988	0.0179	0.00032
0.60%	0.3055	0.6243	-0.1365	0.0868	0.0411	0.0607	0.0651	-0.1129	0.0201	0.00040
0.70%	0.4365	0.5660	-0.1719	0.0966	0.0564	0.0678	0.0755	-0.1270	0.0224	0.00050
0.80%	0.5020	0.5077	-0.2076	0.1065	0.0716	0.0749	0.0859	-0.1411	0.0249	0.00062
0.90%	0.5675	0.4494	-0.2433	0.1165	0.0868	0.0821	0.0964	-0.1553	0.0276	0.00076
1.00%	0.6330	0.3911	-0.2789	0.1262	0.1021	0.0892	0.1068	-0.1694	0.0303	0.00092



Efficient Frontier